In [1]:
```
import numpy as np
import pandas as pd

import matplotlib.pyplot as plt
%matplotlib inline
import matplotlib.gridspec as gridspec
from matplotlib.gridspec import GridSpec

import seaborn as sns

#Plotly
#Plotly
import plotly.express as px
import plotly.graph_objs as go
import plotly.figure_factory as ff

#Some styling
sns.set_style("darkgrid")
plt.style.use("fivethirtyeight")

import plotly.io as pio
pio.templates.default = "gridon"

#Subplots
from plotly.subplots import make_subplots
```

In [2]:
```
df = pd.read_csv("WA_Fn-UseC_-HR-Employee-Attrition.csv")
```

In [3]:
```
#First 5 rows of our dataset
df.head()
```

Out[3]:

	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education	Educ
0	41	Yes	Travel_Rarely	1102	Sales	1	2	Life
1	49	No	Travel_Frequently	279	Research & Development	8	1	Life
2	37	Yes	Travel_Rarely	1373	Research & Development	2	2	
3	33	No	Travel_Frequently	1392	Research & Development	3	4	Life
4	27	No	Travel_Rarely	591	Research & Development	2	1	

5 rows × 35 columns

In [4]:
```
df.shape
```

Out[4]: (1470, 35)

our dataset consists of **1470 columns** and **35 rows**

In [5]:
```
#Columns in our dataset
df.columns
```

Out[5]:
```
Index(['Age', 'Attrition', 'BusinessTravel', 'DailyRate', 'Department',
       'DistanceFromHome', 'Education', 'EducationField', 'EmployeeCount',
       'EmployeeNumber', 'EnvironmentSatisfaction', 'Gender', 'HourlyRate',
       'JobInvolvement', 'JobLevel', 'JobRole', 'JobSatisfaction',
       'MaritalStatus', 'MonthlyIncome', 'MonthlyRate', 'NumCompaniesWorked',
       'Over18', 'OverTime', 'PercentSalaryHike', 'PerformanceRating',
       'RelationshipSatisfaction', 'StandardHours', 'StockOptionLevel',
       'TotalWorkingYears', 'TrainingTimesLastYear', 'WorkLifeBalance',
       'YearsAtCompany', 'YearsInCurrentRole', 'YearsSinceLastPromotion',
       'YearsWithCurrManager'],
      dtype='object')
```

In [6]:
```
df.describe().T
```

Out[6]:

	count	mean	std	min	25%	50%	75%
Age	1470.0	36.923810	9.135373	18.0	30.00	36.0	43.00
DailyRate	1470.0	802.485714	403.509100	102.0	465.00	802.0	1157.00
DistanceFromHome	1470.0	9.192517	8.106864	1.0	2.00	7.0	14.00
Education	1470.0	2.912925	1.024165	1.0	2.00	3.0	4.00
EmployeeCount	1470.0	1.000000	0.000000	1.0	1.00	1.0	1.00
EmployeeNumber	1470.0	1024.865306	602.024335	1.0	491.25	1020.5	1555.75
EnvironmentSatisfaction	1470.0	2.721769	1.093082	1.0	2.00	3.0	4.00
HourlyRate	1470.0	65.891156	20.329428	30.0	48.00	66.0	83.75
JobInvolvement	1470.0	2.729932	0.711561	1.0	2.00	3.0	3.00
JobLevel	1470.0	2.063946	1.106940	1.0	1.00	2.0	3.00
JobSatisfaction	1470.0	2.728571	1.102846	1.0	2.00	3.0	4.00
MonthlyIncome	1470.0	6502.931293	4707.956783	1009.0	2911.00	4919.0	8379.00
MonthlyRate	1470.0	14313.103401	7117.786044	2094.0	8047.00	14235.5	20461.50
NumCompaniesWorked	1470.0	2.693197	2.498009	0.0	1.00	2.0	4.00
PercentSalaryHike	1470.0	15.209524	3.659938	11.0	12.00	14.0	18.00
PerformanceRating	1470.0	3.153741	0.360824	3.0	3.00	3.0	3.00
RelationshipSatisfaction	1470.0	2.712245	1.081209	1.0	2.00	3.0	4.00
StandardHours	1470.0	80.000000	0.000000	80.0	80.00	80.0	80.00
StockOptionLevel	1470.0	0.793878	0.852077	0.0	0.00	1.0	1.00
TotalWorkingYears	1470.0	11.279592	7.780782	0.0	6.00	10.0	15.00
TrainingTimesLastYear	1470.0	2.799320	1.289271	0.0	2.00	3.0	3.00
WorkLifeBalance	1470.0	2.761224	0.706476	1.0	2.00	3.0	3.00
YearsAtCompany	1470.0	7.008163	6.126525	0.0	3.00	5.0	9.00
YearsInCurrentRole	1470.0	4.229252	3.623137	0.0	2.00	3.0	7.00
YearsSinceLastPromotion	1470.0	2.187755	3.222430	0.0	0.00	1.0	3.00
YearsWithCurrManager	1470.0	4.123129	3.568136	0.0	2.00	3.0	7.00

In [7]:
```
#Information of dataset
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1470 entries, 0 to 1469
Data columns (total 35 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Age                       1470 non-null   int64
 1   Attrition                 1470 non-null   object
 2   BusinessTravel            1470 non-null   object
 3   DailyRate                 1470 non-null   int64
 4   Department                1470 non-null   object
 5   DistanceFromHome          1470 non-null   int64
 6   Education                 1470 non-null   int64
 7   EducationField            1470 non-null   object
 8   EmployeeCount             1470 non-null   int64
 9   EmployeeNumber            1470 non-null   int64
 10  EnvironmentSatisfaction   1470 non-null   int64
 11  Gender                    1470 non-null   object
 12  HourlyRate                1470 non-null   int64
 13  JobInvolvement            1470 non-null   int64
 14  JobLevel                  1470 non-null   int64
 15  JobRole                   1470 non-null   object
 16  JobSatisfaction           1470 non-null   int64
 17  MaritalStatus             1470 non-null   object
 18  MonthlyIncome             1470 non-null   int64
 19  MonthlyRate               1470 non-null   int64
 20  NumCompaniesWorked        1470 non-null   int64
 21  Over18                    1470 non-null   object
 22  OverTime                  1470 non-null   object
 23  PercentSalaryHike         1470 non-null   int64
 24  PerformanceRating         1470 non-null   int64
 25  RelationshipSatisfaction  1470 non-null   int64
 26  StandardHours             1470 non-null   int64
 27  StockOptionLevel          1470 non-null   int64
 28  TotalWorkingYears         1470 non-null   int64
 29  TrainingTimesLastYear     1470 non-null   int64
 30  WorkLifeBalance           1470 non-null   int64
 31  YearsAtCompany            1470 non-null   int64
 32  YearsInCurrentRole        1470 non-null   int64
 33  YearsSinceLastPromotion   1470 non-null   int64
 34  YearsWithCurrManager      1470 non-null   int64
dtypes: int64(26), object(9)
memory usage: 402.1+ KB
```

Dataset Data Types Overview

- Our dataset contains two main data types:

1. **Integer** (`int`)
2. **Object** (`object`)

- **Numerical features: 26**

- **Categorical features: 9**
- Some features should be of object data type
- like `Education` , `EnvironmentSatisfaction` , `JobInvolvement` , `JobSatisfaction` , `PerformanceRating` , `RelationshipSatisfaction` , `WorkLifeBalance`

In [8]:
```
df.skew()
```

```
C:\Users\alaqsa\AppData\Local\Temp\ipykernel_13972\1665899112.py:1: FutureWar
ning: The default value of numeric_only in DataFrame.skew is deprecated. In a
future version, it will default to False. In addition, specifying 'numeric_on
ly=None' is deprecated. Select only valid columns or specify the value of num
eric_only to silence this warning.
  df.skew()
```

Out[8]:
```
Age                         0.413286
DailyRate                  -0.003519
DistanceFromHome            0.958118
Education                  -0.289681
EmployeeCount               0.000000
EmployeeNumber              0.016574
EnvironmentSatisfaction    -0.321654
HourlyRate                 -0.032311
JobInvolvement             -0.498419
JobLevel                    1.025401
JobSatisfaction            -0.329672
MonthlyIncome               1.369817
MonthlyRate                 0.018578
NumCompaniesWorked          1.026471
PercentSalaryHike           0.821128
PerformanceRating           1.921883
RelationshipSatisfaction   -0.302828
StandardHours               0.000000
StockOptionLevel            0.968980
TotalWorkingYears           1.117172
TrainingTimesLastYear       0.553124
WorkLifeBalance            -0.552480
YearsAtCompany              1.764529
YearsInCurrentRole          0.917363
YearsSinceLastPromotion     1.984290
YearsWithCurrManager        0.833451
dtype: float64
```

In [9]:
```
features_to_change = [ 'Education' , 'EnvironmentSatisfaction', 'JobInvolvemen
'StockOptionLevel', 'JobLevel', 'BusinessTravel','Department', 'EducationField

int_to_object = ['Education' , 'EnvironmentSatisfaction', 'JobInvolvement', 'J

for feature in int_to_object:
    df[feature] = df[feature].astype(object)
```

Feature to be examined for potential dropping

In [10]:
```
df['EmployeeCount'].value_counts()
```

Out[10]:
```
1    1470
Name: EmployeeCount, dtype: int64
```

In [11]:
```
df['Over18'].value_counts()
```

Out[11]:
```
Y    1470
Name: Over18, dtype: int64
```

In [12]:
```
df['EmployeeNumber'].value_counts()
```

Out[12]:
```
1       1
1391    1
1389    1
1387    1
1383    1
       ..
659     1
657     1
656     1
655     1
2068    1
Name: EmployeeNumber, Length: 1470, dtype: int64
```

- This feature looks like an ID feature

In [13]:
```
df['StandardHours'].value_counts()
```

Out[13]:
```
80    1470
Name: StandardHours, dtype: int64
```

- I think none of the above features added any value, so I will drop them.

In [14]:
```
features_to_drop=['StandardHours', 'EmployeeNumber', 'Over18', 'EmployeeCount'

for feature in features_to_drop :
    df.drop(columns=[feature],axis=1,inplace=True)
```

Null values

In [15]:
```
df.isna().sum().sum()
```

Out[15]:
```
0
```

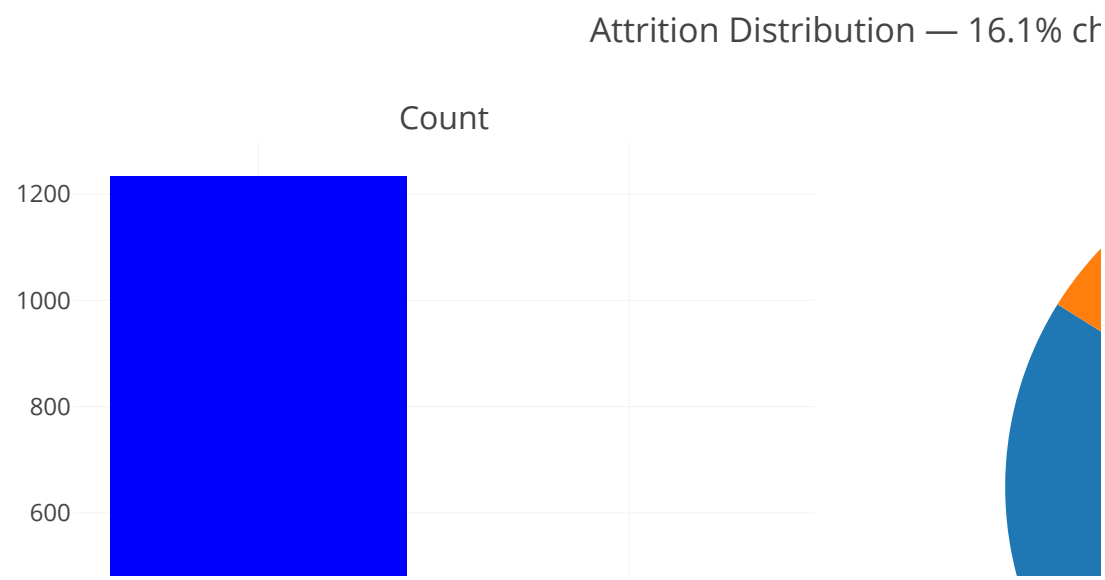
Target Feature

In [16]:
```
df['Attrition'].describe().T
```

Out[16]:
```
count     1470
unique       2
top         No
freq      1233
Name: Attrition, dtype: object
```

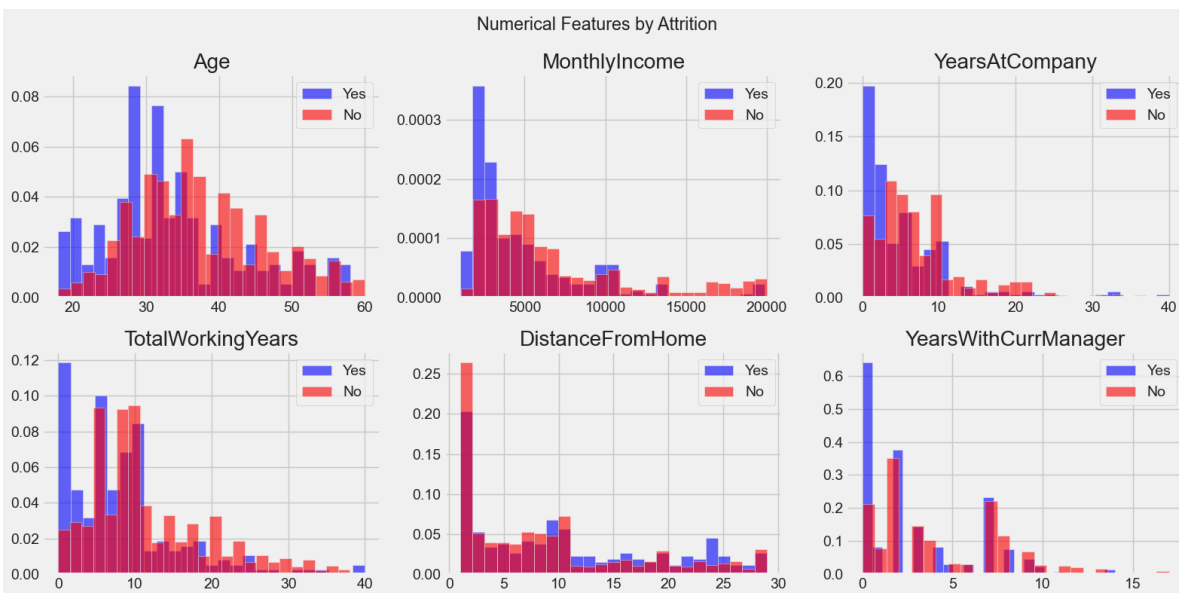
In [26]:
```
# Attrition rate + pie
attrition_counts = df['Attrition'].value_counts()
attrition_rate = round(attrition_counts['Yes'] / len(df) * 100, 1)
print(f"Attrition Rate: {attrition_rate}%")

fig = make_subplots(rows=1, cols=2,
                    subplot_titles=("Count", "Proportion"),
                    specs=[[{"type": "bar"}, {"type": "pie"}]])

fig.add_trace(go.Bar(x=attrition_counts.index, y=attrition_counts.values,
                     marker_color=['blue', 'red']), row=1, col=1)
fig.add_trace(go.Pie(labels=attrition_counts.index, values=attrition_counts.va
                     hole=0.4), row=1, col=2)
fig.update_layout(title=f"Attrition Distribution — {attrition_rate}% churned",
fig.show()
```

```
Attrition Rate: 16.1%
```



- The dataset is clearly **imbalanced** — **83.9%** of employees stayed while only **16.1%** left, This imbalance can bias model performance, as a naive classifier could just predict "No" and still get 84% accuracy.
- Therefore, handling class imbalance (using SMOTE or class weighting) is necessary for reliable modeling.

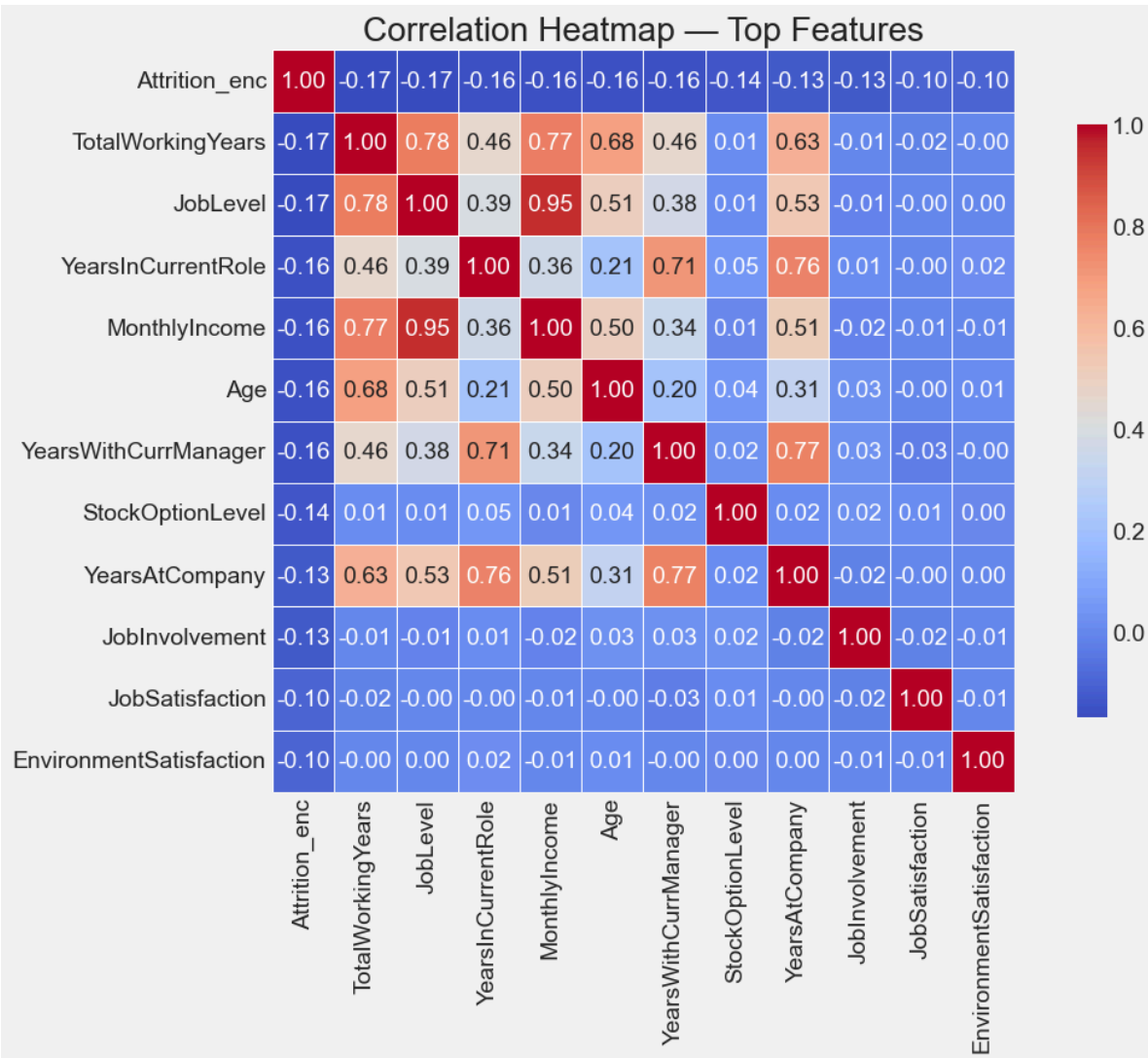
In [27]:
```
num_cols = ['Age', 'MonthlyIncome', 'YearsAtCompany', 'TotalWorkingYears',
            'DistanceFromHome', 'YearsWithCurrManager']

fig, axes = plt.subplots(2, 3, figsize=(16, 8))
axes = axes.flatten()

for i, col in enumerate(num_cols):
    for label, color in zip(['Yes', 'No'], ['blue', 'red']):
        subset = df[df['Attrition'] == label][col]
        axes[i].hist(subset, bins=25, alpha=0.6, color=color, label=label, den
    axes[i].set_title(col)
    axes[i].legend()

plt.suptitle("Numerical Features by Attrition")
plt.tight_layout()
plt.show()
```



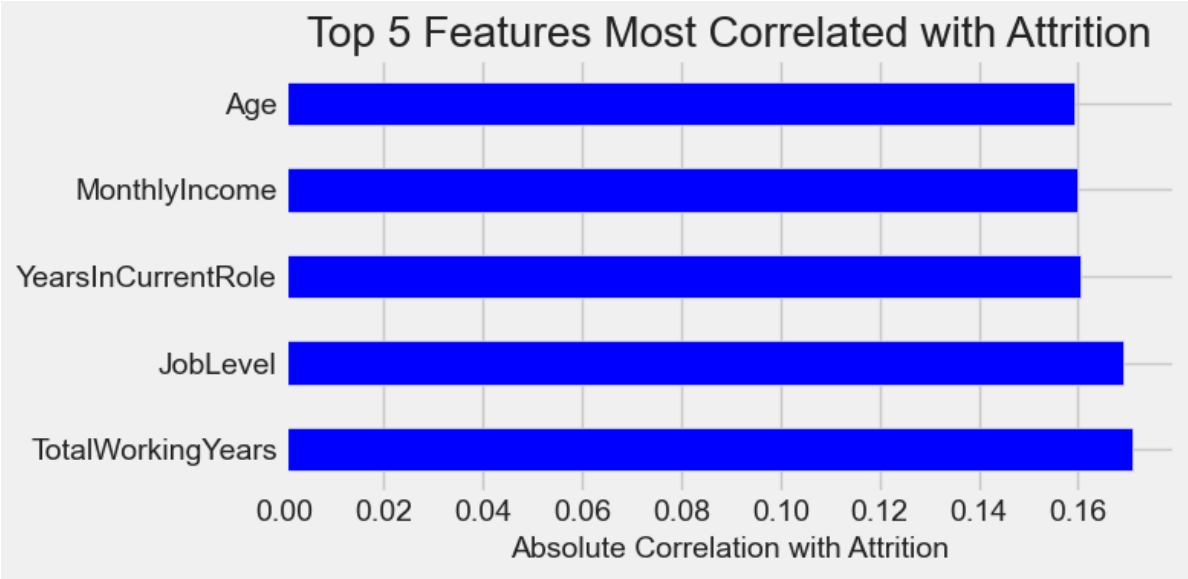
- **Age**: Employees who left are mostly younger **(peak around 25–30)**, while those who stayed are more spread across **30–45. Early career employees are higher risk.**
- **MonthlyIncome**: Most employees who **left** earn **lower salaries (below $5k)**. Employees with **higher income rarely leave**, so income is an important factor for retention.
- **YearsAtCompany**: Employees who **left** usually have **short tenure**, mostly less than **5 years**. The first **1–3 years** seem to be the most important period.
- **TotalWorkingYears**: **Less experienced** employees are more likely to leave. Attrition **decreases a lot after 10+ years** of experience.
- **DistanceFromHome**: Employees who left tend to live a bit farther from work, but the effect is not very strong.

- **YearsWithCurrManager**: Employees who left spent **less time with their manager**, which suggests that having a stable manager helps keep employees

In [28]:
```
# Encode target for correlation
df_corr = df.copy()
df_corr['Attrition_enc'] = (df['Attrition'] == 'Yes').astype(int)

num_df = df_corr.select_dtypes(include='number')
corr = num_df.corr()

# Show only top features correlated with Attrition
top_corr_features = corr['Attrition_enc'].abs().sort_values(ascending=False).h
corr_top = num_df[top_corr_features].corr()

plt.figure(figsize=(12, 9))
sns.heatmap(corr_top, annot=True, fmt=".2f", cmap="coolwarm",
            linewidths=0.5, square=True, cbar_kws={"shrink": 0.8})
plt.title("Correlation Heatmap — Top Features")
plt.tight_layout()
plt.show()
```

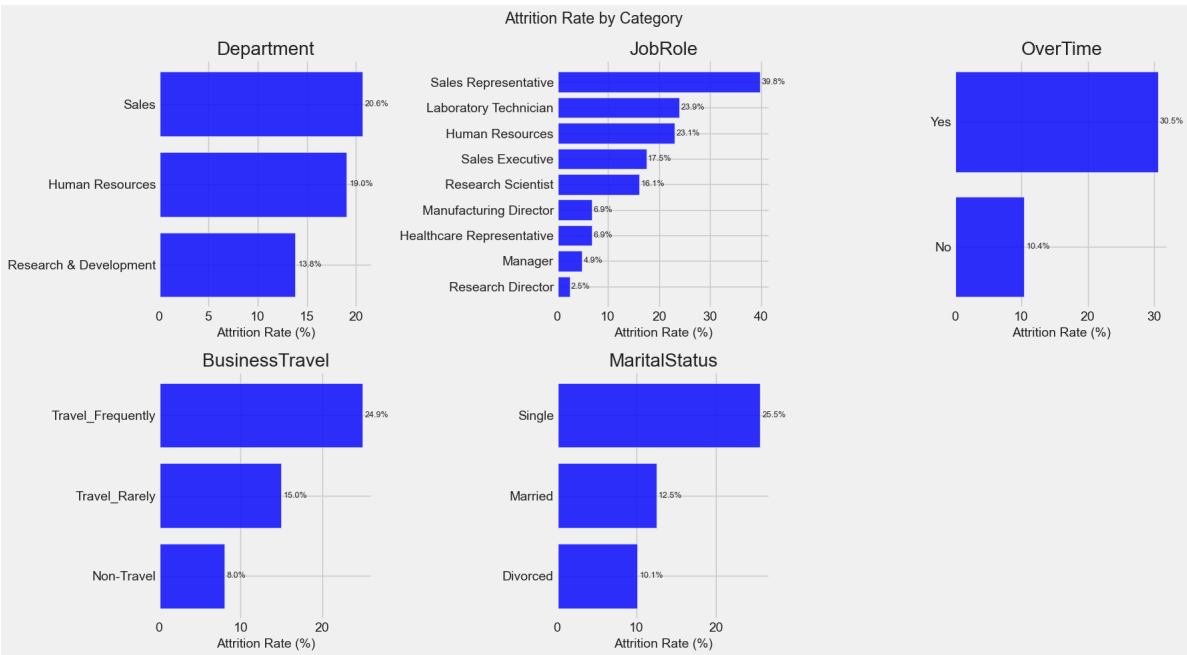


In [30]:
```
top5 = corr['Attrition_enc'].drop('Attrition_enc').abs().nlargest(5)

plt.figure(figsize=(8, 4))
top5.plot(kind='barh', color='blue')
plt.xlabel("Absolute Correlation with Attrition")
plt.title("Top 5 Features Most Correlated with Attrition")
plt.tight_layout()
plt.show()
```



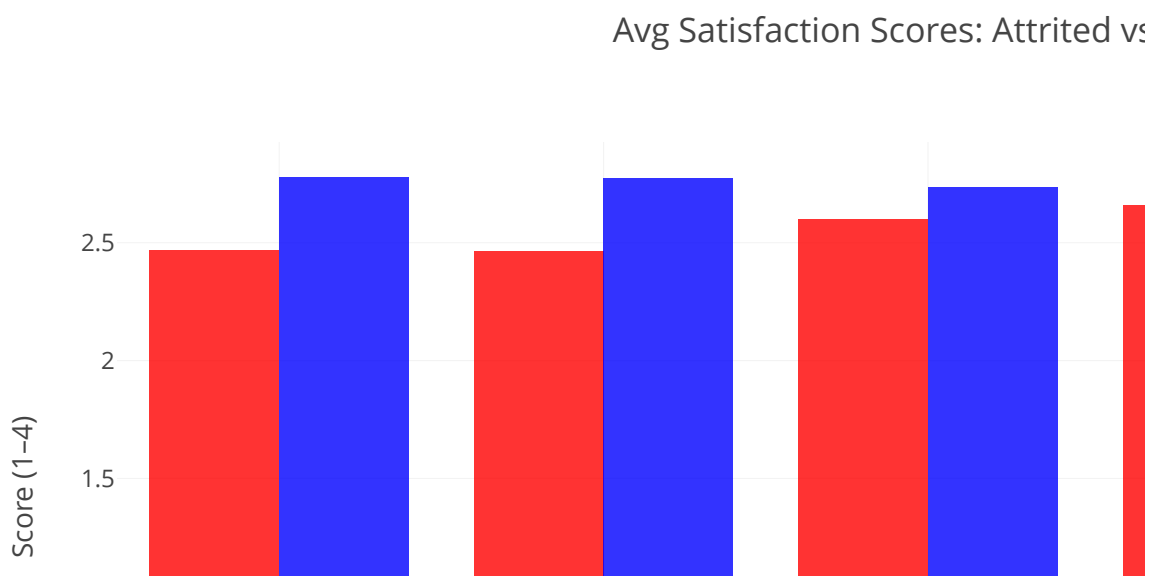
- All top 5 features (TotalWorkingYears, JobLevel, YearsInCurrentRole, MonthlyIncome, Age) move in the same direction, as they all represent experience and seniority. This shows a clear pattern: **junior, less experienced, and lower-paid employees are more likely to leave**.
- The correlations are moderate (around 0.16–0.17), which means **no single feature is the main driver**. Instead, attrition is influenced by a combination of factors rather than one factor alone.

In [31]:
```
cat_cols = ['Department', 'JobRole', 'OverTime', 'BusinessTravel', 'MaritalSta

fig, axes = plt.subplots(2, 3, figsize=(18, 10))
axes = axes.flatten()

for i, col in enumerate(cat_cols):
    rate = df.groupby(col)['Attrition'].apply(lambda x: (x == 'Yes').mean() *
    axes[i].barh(rate.index, rate.values, color='blue', alpha=0.8)
    axes[i].set_xlabel("Attrition Rate (%)")
    axes[i].set_title(col)
    for j, v in enumerate(rate.values):
        axes[i].text(v + 0.3, j, f"{v:.1f}%", va='center', fontsize=9)

axes[-1].set_visible(False)
plt.suptitle("Attrition Rate by Category")
plt.tight_layout()
plt.show()
```



- **OverTime**: Employees who work overtime have **almost 3× higher attrition (30.5% vs 10.4%)**, making it one of the strongest factors in the dataset.
- **JobRole**: **Sales Representatives have the highest attrition (39.8%)**, followed by Lab Technicians (23.9%) and HR (23.1%). **Research Directors have the lowest rate at 2.5%.**
- **BusinessTravel**: Employees who travel frequently have higher attrition (24.9%) compared to non-travelers (8%), showing that travel affects retention.
- **MaritalStatus**: **Single employees leave more often (25.5%) compared to divorced employees (10.1%),** possibly due to fewer personal commitments.
- **Department:** The Sales department has the highest attrition (20.6%), which matches the high attrition seen in Sales roles.

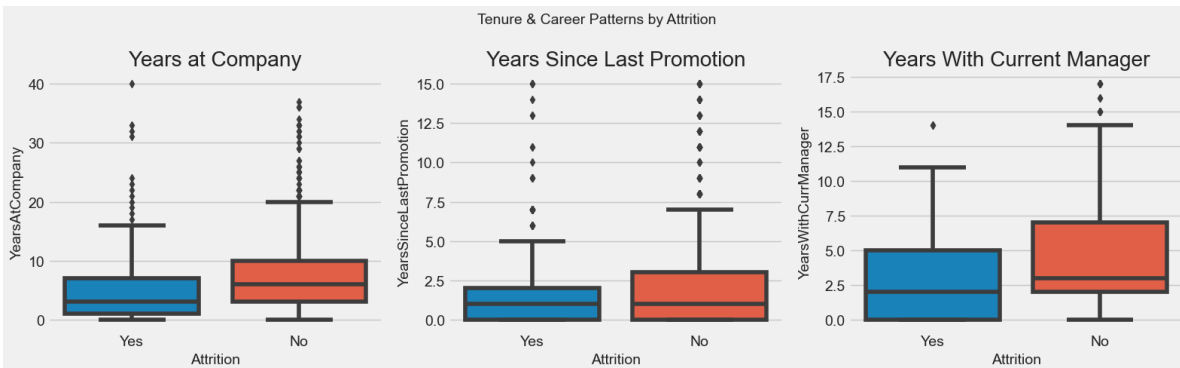
In [32]:
```
# Average satisfaction scores: Attrited vs Stayed
sat_cols = ['JobSatisfaction', 'EnvironmentSatisfaction',
            'RelationshipSatisfaction', 'WorkLifeBalance', 'JobInvolvement']

# Convert back to int for mean calculation
for c in sat_cols:
    df[c] = df[c].astype(int)

sat_means = df.groupby('Attrition')[sat_cols].mean()

fig = go.Figure()
for label, color in zip(['Yes', 'No'], ['red', 'blue']):
    fig.add_trace(go.Bar(name=label, x=sat_cols, y=sat_means.loc[label],
                         marker_color=color, opacity=0.8))

fig.update_layout(barmode='group',
                  title="Avg Satisfaction Scores: Attrited vs Stayed",
                  yaxis_title="Score (1–4)", xaxis_title="")
fig.show()
```



- Employees who left have lower scores in all 5 satisfaction measures. The biggest differences appear in JobSatisfaction and EnvironmentSatisfaction, where their scores are clearly lower.

- Even though the differences seem small (about 0.2–0.3 on a 1–4 scale), they are consistent across all metrics. There is no satisfaction area where employees who left scored the same or higher.
- This shows that dissatisfaction is not caused by one specific factor, but is a general feeling across multiple areas.

In [33]:
```
fig, axes = plt.subplots(1, 3, figsize=(16, 5))

# Years at company
sns.boxplot(data=df, x='Attrition', y='YearsAtCompany', ax=axes[0])
axes[0].set_title("Years at Company")

# Years since last promotion
sns.boxplot(data=df, x='Attrition', y='YearsSinceLastPromotion', ax=axes[1])
axes[1].set_title("Years Since Last Promotion")

# Years with current manager
sns.boxplot(data=df, x='Attrition', y='YearsWithCurrManager', ax=axes[2])
axes[2].set_title("Years With Current Manager")


plt.suptitle("Tenure & Career Patterns by Attrition", fontsize=14)
plt.tight_layout()
plt.show()
```



- **Years at Company**: Employees who left stayed for a shorter time (around 3 years) compared to those who stayed (around 7 years). The first few years are the most risky.
- **Years Since Last Promotion**: **Both groups look similar**, with **low median values**. This suggests that **promotion alone is not the main reason employees leave**.
- **Years With Current Manager**: Employees who left spent less time with their manager. **A short relationship with the manager may indicate a new or unstable team**, or that employees leave before building strong connections — both are warning signs for HR.

In [ ]:
```

```